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CS 306: Machine Learning Lab
Evaluation Assignment 3

Instructions: This assignment is for evaluation and marks will be awarded. You need to complete execution by 12 PM. The saved results folder (consisting .csv files, plots, word file) has to be submitted through a Google form, which will be shared by Teaching Assistant.

1. A environmental monitoring agency wants to identify groups of air quality measurements that exhibit similar pollution characteristics. Consider a dataset that consists of measurements of particulate matter and gaseous pollutants from different monitoring stations across a city with the following attributes:

   - PM2.5: Concentration of fine particulate matter in $\mu g/m^3$ (micrograms per cubic meter).
   - PM10: Concentration of particulate matter with a diameter of 10 micrometres or less, in $\mu g/m^3$.
   - NO2: Nitrogen dioxide concentration in $\mu g/m^3$.
   - SO2: Sulphur dioxide concentration in $\mu g/m^3$.

   Task: Apply a suitable clustering algorithm to to identify groups of monitoring stations with similar air quality observations and detect unusual measurements that do not belong to any dense cluster (noise points/outliers).
   Perform the following to complete the above tasks:

   (a) Read the dataset, and extract the pollutant concentration attributes as the input features.

   (b) Standardize the features using Z-score standardization. Write your observation on whether standardization is a necessary step in this task.

   (c) Implement distinct clustering algorithms for the below parameter settings, to identify the clusters, noise points, and the percentage of noise points for each setting separately.

   - eps :0.5, 1.2, 2.1, 0.8, 0.8.
   - $minpts$:3, 3, 3, 5, 7

   Report your findings for all five combination of parameter settings separately.

   (d) Report how the changes in eps and $minpts$ effect cluster formation and noise detection.

   (e) Visualize the clusters identified using a two-dimensional scatter plot of PM2.5 and PM10. Analyze and report your observations on the characteristics of the identified clusters in terms of pollutant concentrations.

   (f) Compute the Silhouette Score for the resultant clusters from 1(c), provided that the number of clusters and evaluated observations satisfy the **condition**: $2 <= K <= N - 1$ where N: total number of patterns/ data points, K: generated clusters that infer the followings:

1. There must be at least two distinct clusters exist.
2. The number of clusters must be less than the number of evaluated observations/ data points.
3. At least one cluster must contain more than one observation.

Hint: Noise points should be filtered out before evaluating the clusters.

(g) Analyze whether Silhouette Score can be computed, around the noise points. Report your findings.

(h) Summarize your observations and write down your result analysis in a word document from the results and plots/graphs.